

## Table of Contents

|                                                                                      |           |
|--------------------------------------------------------------------------------------|-----------|
| <b>ASSIGNMENT   Machine Learning for Metabolomics .....</b>                          | <b>2</b>  |
| <b>MATTHEW SPRIGGS .....</b>                                                         | <b>2</b>  |
| <b>Report .....</b>                                                                  | <b>3</b>  |
| <b>1. Introduction .....</b>                                                         | <b>3</b>  |
| <b>2. Data Preparation &amp; Exploration [15marks].....</b>                          | <b>4</b>  |
| <b>3. Random Forest Classification for Apple Development Stages [35 marks].....</b>  | <b>10</b> |
| <b>4. Random Forest with Multivariate Longitudinal Predictors. ....</b>              | <b>14</b> |
| <b>5. Random Forest Classification for Apple Storage Treatments [35 marks] .....</b> | <b>15</b> |
| <b>6. Discussion and Biological Interpretation .....</b>                             | <b>16</b> |
| <b>7. Bibliography .....</b>                                                         | <b>18</b> |

# ASSIGNMENT | **Machine Learning for Metabolomics**

MATTHEW SPRIGGS

APPLIED BIOINFORMATICS

ESTIMATED WORD COUNT (EXCLUDING FIGURES AND QUESTIONS): ~3,400 WORDS.

# Report

*General note on running scripts for this pipeline. There is a ".env" file in the assignment/scripts directory. Update this file to point to your local directory.*

## 1. Introduction

- 1.1 The UK apple industry is valued at £977 million (Fruit & Vine Magazine, 2024) with approximately 150,000 tonnes sold by UK supermarkets alone in 24/25 (British Apples & Pears, 2025).
- 1.2 Apple quality improves post-harvest until the fruit is fully ripened. This process involves the conversion of organic acids and starches into sugars and other flavor components, after which point quality declines. During the decline, the fruits firmness and flavor begin to deteriorate — in a process known as senescence — until such a point the fruit is no longer economically viable (Amaro et al., 2022).
- 1.3 In the UK, domestic apple production has significantly declined and the country has built a reliance on importing apple from climate-vulnerable countries (Scheelbeek et al., 2020). It is, therefore, important for UK producers to invest in research and resources to extend the shelf life of locally cultivated apples in alignment with the UK government's objective of self-sufficiency.
- 1.4 The storage of apples for several months is a well-established practice (Büchele et al., 2024). Over the past decade, an intensive effort has been made to optimize the storage conditions to maximize quality — and economic viability. Controlled atmosphere (CA) and Dynamic controlled atmosphere (DCA) are two such technologies.
- 1.5 Controlled atmosphere storage generally consists of increasing the carbon dioxide concentration and decreasing the oxygen concentrations to the effect of reducing metabolic activity and delaying senescence (Yahia & Woolf, 2011).
- 1.6 Dynamic controlled atmosphere can be broadly defined as a system that allows for the customisation of oxygen levels both at the beginning and during storage (Mditshwa et al., 2018). Such systems employ a variety of sensors (chlorophyll fluorescence, respiration quotient, ethanol, and carbon dioxide monitoring).
- 1.7 To be able to classify the effectiveness of any treatment intervention robust quantitative methods are important. "This project aim[s] at assessing the physicochemical changes and quality traits of 'Royal Gala' apples stored for a total of 29 weeks under DCA and cold storage over two successive years. In addition, in Year2 the DCA was compared against Controlled Atmosphere (CA), which replicates current industrial storage practices (1 % O<sub>2</sub> and 5 % CO<sub>2</sub>). (Anastasiadi, 2026)"
- 1.8 Sampling was carried out at weeks 0, 5, 11, 17, 23, and 29 across consecutive years. At each sampling point, a range of physicochemical traits was measured, including phenolic compounds, organic acids, sugars, phytohormone levels, firmness, and total soluble solids (Anastasiadi, 2026).
- 1.9 This report aims to produce an algorithm that can classify apple samples into either DCA or CA conditions based on their physicochemical traits. The report further aims to produce an algorithm that can predict storage week and ripening stage for an apple based on its physicochemical traits.

## 2. Data Preparation & Exploration [15marks]

- 2.1 Data preparation and exploration:** The three raw data files were loaded and checked for missing data. DCA Year 2 had missing values for sample DCA2\_S38. This sample was missing values for Fructose, Sorbitol, Glucose, and Sucrose. A decision was made to impute the mean value for these columns, without accounting for storage week or ripening stage — this seemed low risk. DCA Year 1 had 432 missing values, where all predictor variables were missing for samples in week 29. The decision was made to drop these samples from the analysis.
- 2.2 Quality Control:** A discrepancy in column names was identified between DCA Year 1 and Year 2 for the phytohormone 7OH-ABA. It was labelled “7OH-ABA” in Year 1 and “7-OH-ABA” in Year 2. The column was renamed to “7-OH-ABA” in both datasets for consistency.
- 2.3 Duplicated Rows:** For week 0, seventeen identical rows were found when DCA Year 2 and CA Year 2 were combined. In addition, rows DCA2\_S18 and CA-S18 were near-duplicates. It was not possible to determine which was correct; therefore, all rows from these datasets for week 0 were removed from downstream analysis.
- 2.4 Principal Component Analysis (PCA) — parameter justification:** A series of principal component analyses were conducted on the individual datasets to explore patterns in the data. For the PCAs, different data transformations were tested to assess the effect of scaling and centring. Detailed interpretation of these PCAs can be found in 01\_data\_prep\_QC.R; however, to summarise Table 1, “centred and scaled” PCAs showed a modest proportion of variance explained by the first two principal components (45%), with an acceptable standard deviation (16.21). Whilst “scaling only” appeared to account for 86% of the variation, without centring the data this variation is still predominantly influenced by the mean structure. Given that predictor variables have different units (e.g., Glucose is expressed in mg/g dry weight from 30–157 mg/g, whilst ABA ranges from 6.3–185 ng/g), scaling alone without centring would be inappropriate. Results are shown for CA Year 2 only, but the same trend was evident across all datasets. Therefore, the decision was taken to scale and centre the data for PCAs.

Table 1: PCA normalization tuning for CA Year 2 data.

| Dataset                | Condition           | explained variance for PC1 + PC2 (%) | stDev for PC1 + PC2 |
|------------------------|---------------------|--------------------------------------|---------------------|
| ca_year_2_pca          | Centered and scaled | 45.05                                | 16.21               |
| ca_year_2_pca_scaled   | Scaled only         | 86.82                                | 31.25               |
| ca_year_2_pca_centered | Centered only       | 89.26                                | >16,000             |
| ca_year_2_pca_none     | Raw                 | 99.10                                | >41,000             |

- 2.5 Principal Component Analysis (PCA) — data exploration:** Biplots from the R package mixOmics were used to further explore the PCA results (Rohart et al., 2017). Factors for storage week and ripening stage were overlaid onto principal components 1 and 2 (Figure 1, Figure 2, Figure 3).

- 2.6 Figure 1 depicts the PCA models for the three individual datasets, split by storage week. PC1 had greater explanatory power in CA Year 2 and DCA Year 2. Notably, these two datasets include data for week 29, whereas DCA Year 1 does not; however, DCA Year 1 includes data for week 0, whereas CA Year 2 and DCA Year 2 do not. This suggests that late-stage profiles may be more informative when classifying apples.
- 2.7 In both Figure 1 [1] and [3], week 29 (magenta) points cluster together, and week 0 (blue) points also cluster together. This trend is less evident in [2], although week 23 (teal) shows a distinct grouping.
- 2.8 Figure 2 depicts the PCA models for the three individual datasets, split by ripening stage. The same trend applies, with the explanatory power of DCA Year 1 greatly reduced by the missing data. Across the plots, early (blue) and late (orange) points cluster into groups (with differing degrees of variability), while mid (grey) points scatter less consistently.
- 2.9 To investigate batch effects, the data from DCA Year 1 and Year 2 were merged. An apparent separation of groups on the y-axis is shown in Figure 3, and is made clear in Figure 4. Figure 4: PC2 scores over storage week [1] and ripening stage [2] split by year. This implies that the physicochemical parameters between batches need be taken into account when building prediction or classification models.



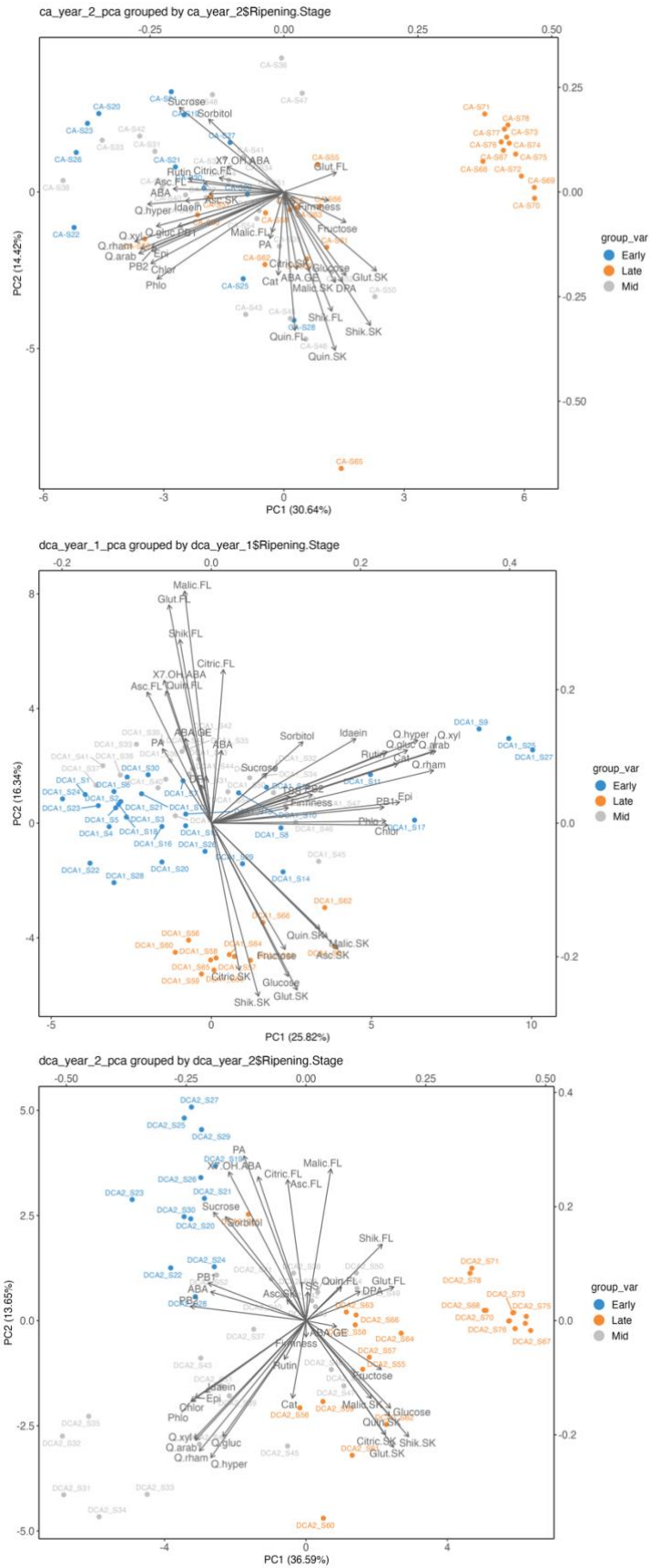


Figure 2: PCA biplots from mixOmics depicting CA Year 2 [1], DCA Year 1 [2] and DCA Year 2 [3] grouped by ripening stage.

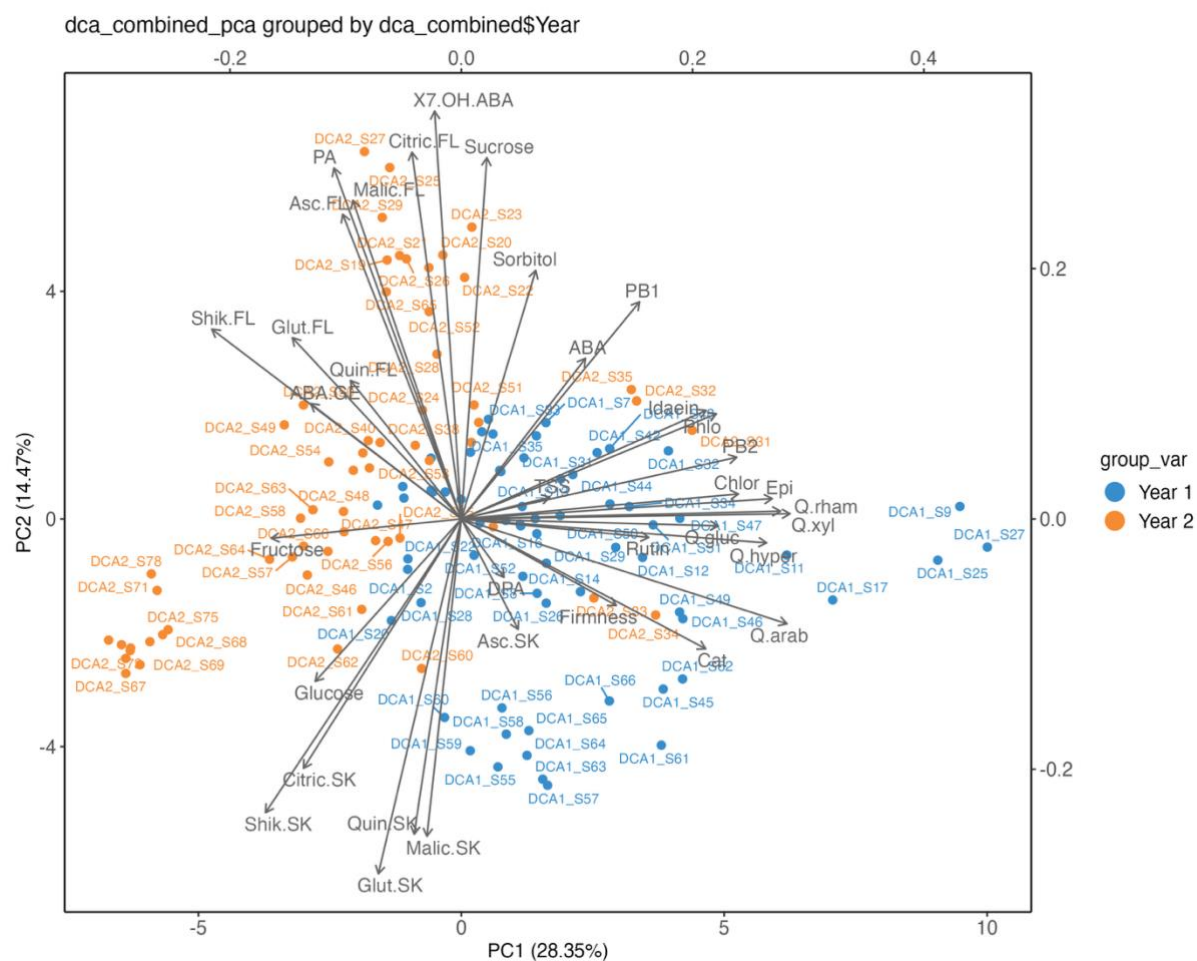


Figure 3: PCA biplots from mixOmics depicting DCA Year 1 and DCA Year 2 combined, grouped by year.

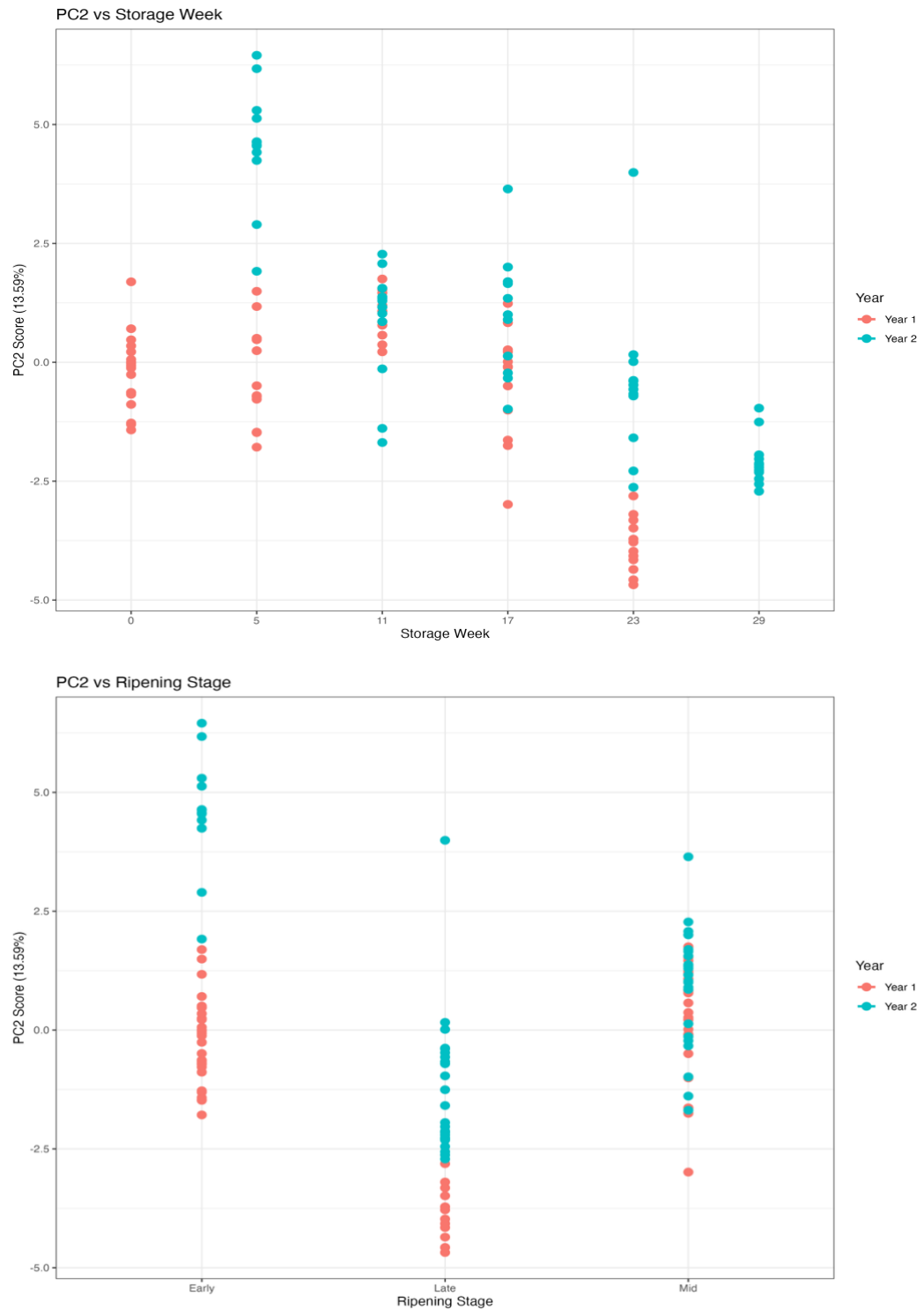


Figure 4: PC2 scores over storage week [1] and ripening stage [2] split by year.

### 3. Random Forest Classification for Apple Development Stages [35 marks]

- 3.1 The physicochemical data from the DCA trial were used to create classification models to predict storage week or ripening stage. These models used DCA Year 1 data for training and DCA Year 2 data for testing.
- 3.2 The pipeline used is 01\_objective\_storage\_ripe\_classification\_models.Rmd.
- 3.3 Missing data and mislabelled column names was handled as per section 1.3.
- 3.4 The proportion of storage week in the training data was reasonably well balanced, with a larger proportion of week 0 samples ; however, the overall balance was adequate Table 2.
- 3.5 The proportions of ripening in the training data were less well balanced Table 3. Any model built using this factor shall have to take into account this imbalance.

Table 2: Storage week proportions and frequencies in DCA Year 1 training data.

| Storage Week | Proportion | N  |
|--------------|------------|----|
| 0            | 27.27      | 18 |
| 5            | 18.18      | 12 |
| 11           | 18.18      | 12 |
| 17           | 18.18      | 12 |
| 23           | 18.18      | 12 |

Table 3: Ripening stage proportions and frequencies in DCA Year 1 training data

| Ripening Stage | Proportion | n  |
|----------------|------------|----|
| Early          | 45.45      | 30 |
| Mid            | 36.36      | 24 |
| Late           | 18.18      | 12 |

- 3.6 **Storage Weeks:** A cross validation method was used to train the model, first creating five groups, each with a balanced mix of storage weeks. This approach has limitations, discussed below.

3.6.1 *Note: the advice was to use createTimeSlices() to split the data using variations on below code, with skipping and without skipping. Different initial windows and horizons were tested. However, a successful model was never built, with repeated errors indicating that one or more factor levels in the outcome had no data: '23'. One conclusion is that the dataset may be too small for this approach. Further work could explore methods to increase the effective sample size (e.g., bootstrapping).*

```
createTimeSlices(1:nrow(training_features), initialWindow =  
floor(0.8 * nrow(training_features)), horizon = floor(0.2 *  
nrow(training_features)), fixedWindow = FALSE)
```

- 3.7 The cross-validation did not consider the temporal structure of the data (e.g., that week 0 preceded week 5 and week 11, etc.). However, this approach yielded a model, discussed below.
- 3.8 The trainControl function from caret was used to split data for the for modelling (Kuhn, 2008). A k-fold cross-validation approach was selected with five folds, chosen to reflect the five factor levels in the DCA Year 1 data. An index function was used to ensure each fold has a similar class distribution, reflecting Table 2.
- 3.9 The train function from caret was then used to train the model based on these splits (Kuhn, 2008). A preprocess parameter was used to scale and center the data as discussed in Table 1. TuneGrid was also used as an argument to optimize the hyperparameter mtry, which controls how many predictors are randomly considered in each split.
- 3.10 This model was then tested on DCA Year 2 data using the predict function, ensuring the factors in the test data matched that in the training data. This was an important consideration, since the training data was not exposed to any week 29 samples.

Table 4: Basic Statistics for Storage.Week cv Random Forest model Trn: DCA Y1 Tst: DCA Y2

| Accuracy | Kappa |
|----------|-------|
| 0.270    | 0.027 |

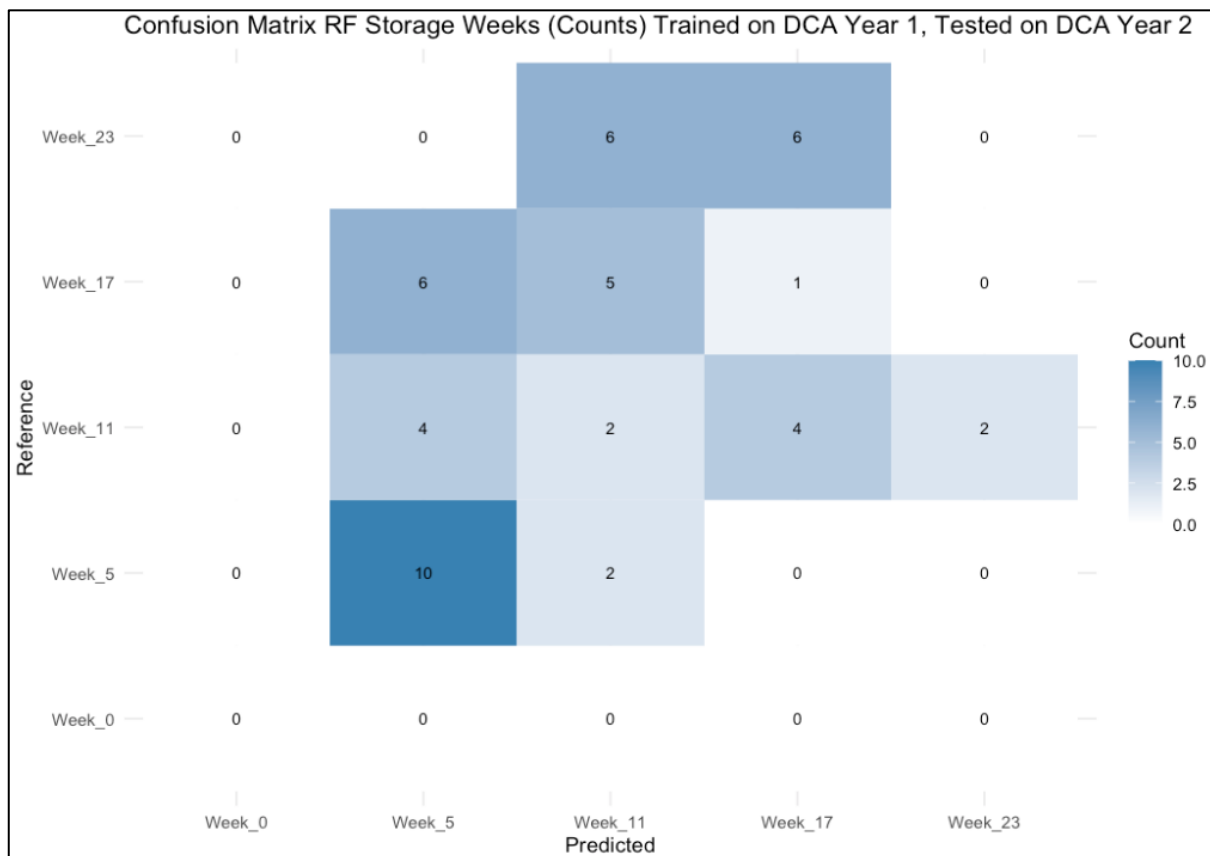


Figure 5: Confusion Matrix for Storage.Week cv Random Forest model tested on DCA Year 2 data, trained on DCA Year 1 data.

- 3.11 Overall, the model was essentially inaccurate. With an accuracy of 27% and a Kappa statistic of 0.02, it was not sufficient to predict storage week based on physicochemical profile.
- 3.12 The model was not able to predict any of the week 29 data. It was not exposed to this factor level during training and, since the data could not be split temporally, it could not make use of the time-series nature of the storage-week factor.
- 3.13 The model was trained on week 0 data, but since there were no week 0 samples in the test set, it — perhaps by chance — did not predict any samples as belonging to this condition.
- 3.14 The process above was repeated, but with Year 2 used to train and optimise the model and Year 1 used as the test set. The pipeline was repeated using DCA Year 2 to train the model and DCA Year 1 to test the model.

Table 5: Basic Statistics for Storage.Week cv Random Forest model Trn: DCA Y2 Tst: DCA Y1

| Accuracy | Kappa |
|----------|-------|
| 0.250    | 0     |

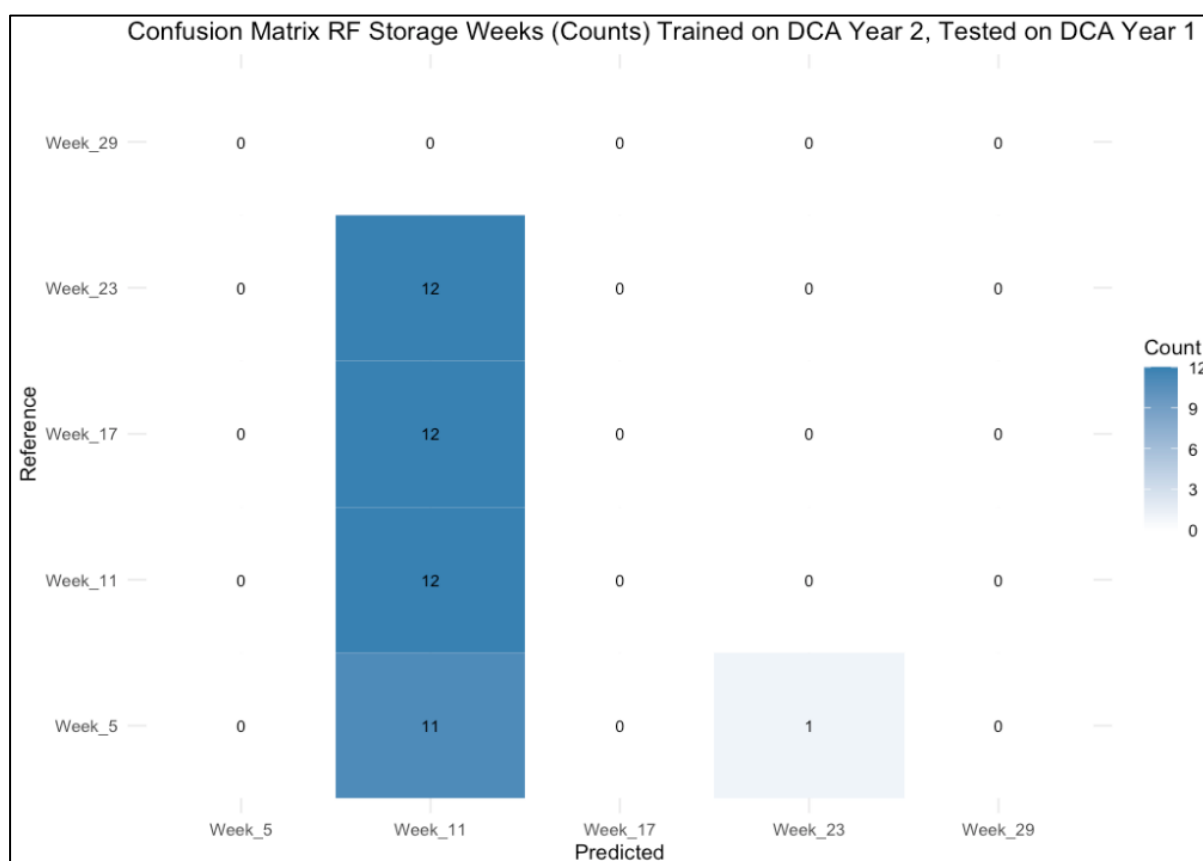


Figure 6: Confusion Matrix for Storage.Week cv Random Forest model tested on DCA Year 1 data, trained on DCA Year 2 data.

- 3.15 The model trained on DCA Year 2 data did not perform as well as the model trained on DCA Year 1 (Table 5), with an accuracy of 25% and a Kappa statistic of 0. This model is effectively worse than random. The model predicted that most samples belonged to week

11 Figure 6. This may reflect an averaging effect, with predictions tending towards the overall mean profile across weeks. The inability to split the data temporally is limiting these models, since each week is treated as an independent group rather than week 5 preceding week 11, week 17, etc. One constraint with this dataset is that each apple is an independent measurement: not measuring the same apple over time, but different apples across the same time period (i.e., not repeated measures).

3.16 **Ripening stage:** The same pipeline was run for ripening stage.

Table 6: Basic Statistics for Ripening.Stage cv Random Forest model Trn: DCA Y1 Tst: DCA Y2

| Accuracy | Kappa |
|----------|-------|
| 0.233    | 0.008 |

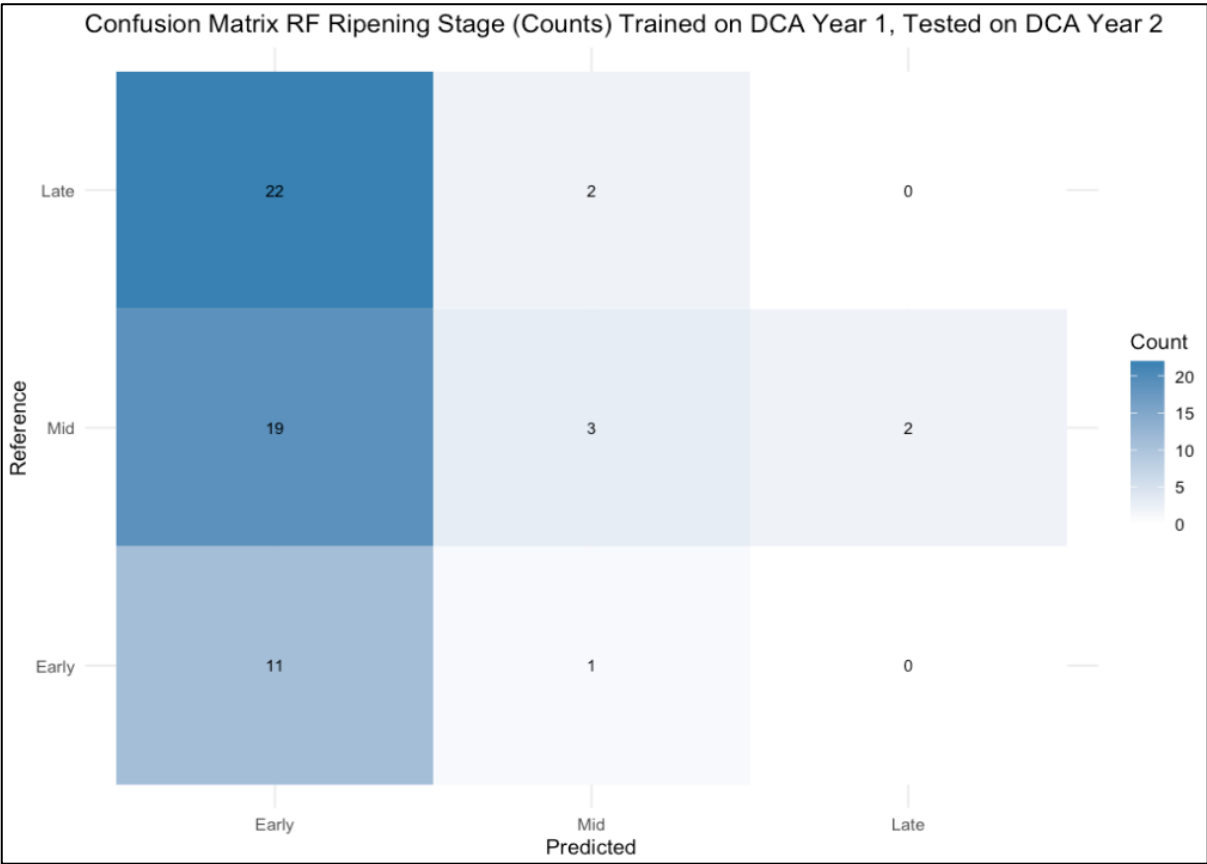


Figure 7:Confusion Matrix for Ripening.Stage cv Random Forest model tested on DCA Year 2 data, trained on DCA Year 1 data

3.17 This model also did not accurately predict the ripening stage of apples based on their physicochemical properties. The model correctly predicted most apples as early, given the class distribution in the training data (Table 3), where 45% of samples are classified as early. This model likely reflects the class imbalance.

3.18 The process above was repeated, but with Year 2 used to train and optimise the model and Year 1 used as the test set. The pipeline was repeated using DCA Year 2 to train the model and DCA Year 1 to test the model.

Table 7: Basic Statistics for Ripening.Stage cv Random Forest model Trn: DCA Y2 Tst: DCA Y1

| Accuracy | Kappa |
|----------|-------|
| 0.363    | 0.017 |



Figure 8: Confusion Matrix for Ripening.Stage cv Random Forest model tested on DCA Year 1 data, trained on DCA Year 2 data

3.19 As with the storage-week model, the model produced largely predicts the average outcome. It predicts most apples as belonging to the mid category. As such, this model cannot be used to make accurate predictions.

#### 4. Random Forest with Multivariate Longitudinal Predictors.

4.1 This task was not attempted, as I was unable to treat the data as a time series. I attempted to follow the DynForest tutorial, but frankly, I did not understand it.

## 5. Random Forest Classification for Apple Storage Treatments [35 marks]

**Acknowledgement of AI Assistance:** *This section was developed with substantial assistance from a Large Language Model (LLM). While I have reviewed and verified the code outputs and understand the conceptual workflow, I acknowledge that the code implementation was predominantly LLM-generated. Independent understanding of the technical details remains limited. Please mark accordingly.*

- 5.1 For this classification model, an attempt was made to coerce the data into a meaningful time series, using bootstrapping to increase the available data.
- 5.2 Data from DCA Year 2 and CA Year 2 were merged, and week 0 duplicates were removed.
- 5.3 A summary matrix for CA and DCA was generated, where each week was summarised by the mean of each metabolite column. For each treatment–week combination, the base R function `sample.int` was used to randomly select 12 samples with replacement and compute a bootstrap mean for each metabolite. In effect, 12 samples were used to create one 36-length vector of weekly means, stored as one row in a 6×36 matrix.
- 5.4 This matrix was then looped over so for each metabolite, the 6 consecutive weekly mean values were treated as a time series and used to compute 22 time-series features using the `Rcatch22` library. Feature names were prefixed with the metabolite name (e.g., `Metabolite__Feature`), returning one concatenated feature vector of length 792 per bootstrap replicate.
- 5.5 Next, the data was partitioned into training and test sets using `createDataPartition` from `caret`, and a similar modelling pipeline as in Section 4 was applied.
- 5.6 The code generates a warning: " Warning: These variables have zero variances". Suggesting that a subsection of variables have no influence on the result, and could be removed in a future implementation.

*Table 8: Basic Statistics for storage condition Rcatch22 model.*

| Accuracy | Kappa |
|----------|-------|
| 1.0      | 1.0   |

*Table 9: Confusion matrix for bootstrapped model used to predict storage condition.*

| Actual | Predicted CA | Predicted DCA |
|--------|--------------|---------------|
| CA     | 150          | 0             |
| DCA    | 0            | 150           |

- 5.7 Results for the `Rcatch22` model are shown in Table 8 and Table 9, predicting storage condition from physicochemical profiles. Whilst the model achieves high accuracy, such strong performance may indicate overfitting or data leakage, perhaps due to how the bootstrapping was carried out.
- 5.8 When this model was tested against DCA Year 1, the model should, theoretically, be able to predict these samples as belonging to the DCA condition.

*Table 10: Confusion matrix for bootstrapped model used to predict storage condition, tested on DCA Year 1 data.*

| Actual | Predicted CA | Predicted DCA |
|--------|--------------|---------------|
| CA     | 0            | 0             |
| DCA    | 500          | 0             |

5.9 Table 10 shows that the model was not able to classify DCA Year 1 data as belonging to the DCA condition. Instead, it classified all 500 bootstrap samples as belonging to the CA condition. As shown in Figure 4, there is a considerable batch effect across years, and the current model was trained only on DCA Year 2 and CA Year 2; therefore, it has no awareness of what Year 1 batches look like. This may be skewing the results. Future models may wish to include Year 1 in the training data, whilst maintaining a clear separation between training and test sets.

#### *End of Acknowledgement of AI Assistance.*

## 6. Discussion and Biological Interpretation

- 6.1 The models produced by this report were learning opportunities but yielded no successful results, thereby feature extraction from this data would not be deemed valid. The discussion will focus on consensus of the available literature.
- 6.2 Destructive and nondestructive methodologies exist to estimate the internal properties of fruits. Naturally, nondestructive methods are preferred where possible since destructive methods are generally labor-intensive, time-consuming and require specialized resources (Sabzi et al., 2022). A variety of nondestructive methods exist, such as machine vision, near-infrared (NIR), and visible/short wavelength infrared spectroscopy, Raman spectroscopy, X-ray imaging and nuclear magnetic resonance imaging, as summarised by Sabzi et al., (2022).
- 6.3 Sabzi et al. (2022) developed a classification model to group apples into four ripeness levels based on physiochemical properties, with a classification success rate of 97%. This paper used an nondestructive machine vision system. Using similar methodologies, Cárdenas-Pérez et al. (2017) classify with 100% accuracy apple ripening stages. Physicochemical parameters such as titratable acidity (TA), soluble solid content (SSC), and firmness were used as confirmation of ripening stage.
- 6.4 "Fruit firmness along with total soluble solid and the starch content are among the main quality parameters used by apple growers to determine the optimum harvest date." (Giné-Bordonaba et al., 2019).
- 6.5 Recent studies have shown that abscisic acid (ABA) — in combination with other hormones — plays a role in ethylene availability, triggering fruit ripening (Fernández-Cancelo et al., 2022; Liu et al., 2020). Suggesting ABA levels are an important variable when considering prediction or classification models. Fine tuning the ABA threshold is likely to be cultivar specific, and may vary between batches.
- 6.6 Malic acid is hypothesised to be important respiratory substrate in post-harvest, although malate adds little to the fruit's metabolic substrate supply, as it is continually produced

rather than broken down during development and ripening (Giné-Bordonaba et al., 2019). Malic acid may be a good predictor of storage time (Thewes et al., 2024).

- 6.7 Fructose content is low at the time of harvest and higher after 6 months of storage, as reported by Thewes et al. (2024). Thus, this metric may act as a good predictor for ripening stage. This paper also report higher fructose under CA than DCA storage (Thewes et al., 2024). Inferring this might be a suitable metric to classify storage conditions.
- 6.8 Sorbitol, a sugar alcohol present in apple fruit, decreases over storage under normal atmospheric conditions (Thewes et al., 2024). However, under DCA has been observed to increase (Thewes et al., 2024). Thus higher sorbitol may be indicative for DCA treatment condition.
- 6.9 Finally, when apples break down sugars (glucose/fructose), they make pyruvate, which normally enters the TCA cycle during normal oxygen levels. But when oxygen gets too low, like under DCA storage, the TCA cycle slows or stops and pyruvate starts to accumulate (Thewes et al., 2024). Under anaerobic conditions e.g. when the oxygen concentration falls too low, more lactic acid and ethanol is produced. In DCA conditions toxic acetaldehyde is converted to acetic acid or ethanol, although, which byproduct depended on the type of DCA storage used (Thewes et al., 2024). Thus suggesting that sugar contents may be a reliable indicator of storage condition, but since these are secondary metabolites likely involved in complex pathways they may not be the most reliable indicators of storage condition.
- 6.10 This discussion presents a theoretical overview of physicochemical traits that may, or may not be relevant to predicting storage week or storage conditions. Unfortunately, due to the technical limitations of the models produced in this report a more data driven approach could not be conducted.

## 7. Bibliography

- Amaro, A. L., Beaulieu, J. C., Grimm, C. C., Stein, R. E., & Almeida, D. P. F. (2022). Quality changes and shelf-life prediction model of postharvest apples using partial least squares and artificial neural network analysis. *Food Chemistry*, 394(1), 133526. <https://doi.org/10.1016/j.foodchem.2011.06.052>
- Anastasiadi, M. (2026). *MSc Applied Bioinformatics ML for Metabolomics Assignment Machine Learning for Metabolomics*. <https://topepo.github.io/caret/data-splitting.html>
- British Apples & Pears. (2025). *Apple Sales Summary - British Apples & Pears*. <https://www.britishapplesandpears.co.uk/apple-sales-summary/>
- Büchele, F., Hivare, K., Khera, K., Thewes, F. R., Argenta, L. C., Hoffmann, T. G., Mahajan, P. V., Prange, R. K., Pareek, S., & Neuwald, D. A. (2024). Novel Energy-Saving Strategies in Apple Storage: A Review. *Sustainability* 2024, Vol. 16, Page 1052, 16(3), 1052. <https://doi.org/10.3390/su16031052>
- Cárdenas-Pérez, S., Chanona-Pérez, J., Méndez-Méndez, J. V., Calderón-Domínguez, G., López-Santiago, R., Perea-Flores, M. J., & Arzate-Vázquez, I. (2017). Evaluation of the ripening stages of apple (Golden Delicious) by means of computer vision system. *Biosystems Engineering*, 159, 46–58. <https://doi.org/10.1016/j.biosystemseng.2017.04.009>
- Fernández-Cancelo, P., Muñoz, P., Echeverría, G., Larrigaudière, C., Teixidó, N., Munné-Bosch, S., & Giné-Bordonaba, J. (2022). Ethylene and abscisic acid play a key role in modulating apple ripening after harvest and after cold-storage. *Postharvest Biology and Technology*, 188, 111902. <https://doi.org/10.1016/j.postharvbio.2022.111902>
- Fruit & Vine Magazine. (2024, July 1). *A closer look at consumer behaviour and the future of apples and pears - Fruit & Vine*. <https://www.fruitandvine.co.uk/a-closer-look-at-consumer-behaviour-and-the-future-of-apples-and-pears/>
- Giné-Bordonaba, J., Echeverría, G., Duaigües, E., Bobo, G., & Larrigaudière, C. (2019). A comprehensive study on the main physiological and biochemical changes occurring during growth and on-tree ripening of two apple varieties with different postharvest behaviour. *Plant Physiology and Biochemistry*, 135, 601–610. <https://doi.org/10.1016/j.plaphy.2018.10.035>
- Kuhn, M. (2008). Building Predictive Models in R Using the caret Package. *Journal of Statistical Software*, 28(5), 1–26. <https://doi.org/10.18637/jss.v028.i05>
- Liu, Y., Tang, M., Liu, M., Su, D., Chen, J., Gao, Y., Bouzayen, M., & Li, Z. (2020). The Molecular Regulation of Ethylene in Fruit Ripening. *Small Methods*, 4(8), 1900485. <https://doi.org/10.1002/smt.201900485>
- Mditshwa, A., Fawole, O. A., & Opara, U. L. (2018). Recent developments on dynamic controlled atmosphere storage of apples—A review. *Food Packaging and Shelf Life*, 16, 59–68. <https://doi.org/10.1016/j.fpsl.2018.01.011>
- Rohart, F., Gautier, B., Singh, A., & Lê Cao, K. A. (2017). mixOmics: An R package for ‘omics feature selection and multiple data integration. *PLOS Computational Biology*, 13(11), e1005752. <https://doi.org/10.1371/journal.pcbi.1005752>
- Sabzi, S., Nadimi, M., Abbaspour-Gilandeh, Y., & Paliwal, J. (2022). Non-Destructive Estimation of Physicochemical Properties and Detection of Ripeness Level of Apples Using Machine Vision. *International Journal of Fruit Science*, 22(1), 628–645. <https://doi.org/10.1080/15538362.2022.2092580>
- Scheelbeek, P. F. D., Moss, C., Kastner, T., Alae-Carew, C., Jarmul, S., Green, R., Taylor, A., Haines, A., & Dangour, A. D. (2020). United Kingdom’s fruit and vegetable supply is increasingly dependent on imports from climate-vulnerable producing countries. *Nature Food* 2020 1:11, 1(11), 705–712. <https://doi.org/10.1038/s43016-020-00179-4>
- Thewes, F. R., Büchele, F., Uhlmann, L. O., Lugaresi, A., de Oliveria Neuwald, D. Q., Brackmann, A., Both, V., Wagner, R., & Neuwald, D. A. (2024). Metabolic Profiling of ‘Elstar’ and ‘Nicoter’

Apples: Impact of Storage Time, Dynamic Controlled Atmosphere and 1-MCP Treatment.  
*Horticulturae*, 10(12), 1372. <https://doi.org/10.3390/horticulturae10121372>

Yahia, E. M., & Woolf, A. B. (2011). Avocado ( *Persea americana* Mill.). *Postharvest Biology and Technology of Tropical and Subtropical Fruits*, 125–186e.  
<https://doi.org/10.1533/9780857092762.125>